

Ratio: answers and worked method

Answers to step-by-step practice 1

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £84 in the ratio 3:4. How much does A receive?	Notebooks and folders are in the ratio 4:7. There are 35 folders. How many notebooks are there?	Adults and children are in the ratio 2:5. There are 24 more children than adults. How many people are there altogether?
Keywords & Calculation	Total known: $3 + 4 = 7$ parts. Find 1 part, then A's 3 parts.	35 folders is 7 parts. Find 1 part, then 4 parts for notebooks.	24 more children is $5 - 2 = 3$ parts. Find 1 part, then 7 parts altogether.
Written Method	$£84 \div 7 = £12$ per part. $£12 \times 3 = £36$. A B £84 altogether 1 part = £12	$35 \div 7 = 5$ per part. $5 \times 4 = 20$ notebooks. Notebooks Folders Folders: 35 1 part = 5 items	$24 \div 3 = 8$ per part. $8 \times 7 = 56$ people. Adults Children 3 extra parts = 24 people 1 part = 8 people
Ballpark Check & Math	£36 is less than half of £84. The other share is £48; together they make £84.	4 parts is just over half of 7 parts, so expect just over half of 35. 20 fits.	The total is 7 parts, the difference 3. Expect between 2×24 and 3×24 : 48 to 72. 56 fits.
Final Answer	A receives £36.	20 notebooks.	56 people altogether.

Ratio: check your method

Step-by-step practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 7 parts; $\text{£}84 \div 7 = \text{£}12$ per part; $\text{£}12 \times 3 = \text{£}36$ for A.	Method 35 folders represent 7 parts. $35 \div 7 = 5$ per part; $5 \times 4 = 20$ notebooks.	Method The difference is $5 - 2 = 3$ parts. $24 \div 3 = 8$ per part; $8 \times 7 = 56$ people.
If you got... $\text{£}48$: check whether you used $\text{£}12 \times 4$. That is B's share; A has 3 parts.	If you got... 32 notebooks: check whether you used $35 - (7 - 4)$. The difference of 3 is in parts, not individual items.	If you got... 168 people: check whether you used 24×7 . The 24 represents 3 parts, not 1; divide by 3 first.
Check A: $\text{£}36$; B: $\text{£}48$. $\text{£}36 + \text{£}48 = \text{£}84$, and 36:48 simplifies to 3:4.	Check 20:35 simplifies to 4:7. There should be fewer notebooks than folders.	Check Adults: 16; children: 40. $40 - 16 = 24$; $16 + 40 = 56$.

Ratio: answers and worked method

Answers to step-by-step practice 2

	Thread 1: Guided	Thread 2: Core	Thread 3: Depth
The Full Question	A and B share £96 in the ratio 5:7. How much does B receive?	Juice and water are mixed in the ratio 2:7. There are 18 litres of juice. How many litres of water are needed?	Two lengths of ribbon, A and B, are in the ratio 5:8. B is 21 cm longer than A. What is their total length?
Keywords & Calculation	Total known: $5 + 7 = 12$ parts. Find 1 part, then B's 7 parts.	18 litres of juice is 2 parts. Find 1 part, then 7 parts of water.	21 cm is $8 - 5 = 3$ parts. Find 1 part, then 13 parts altogether.
Written Method	$£96 \div 12 = £8$ per part. $£8 \times 7 = £56$. A B £96 altogether 1 part = £8	$18 \div 2 = 9$ litres per part. $9 \times 7 = 63$ litres. Juice Water Juice: 18 litres 1 part = 9 litres	$21 \div 3 = 7$ cm per part. $7 \times 13 = 91$ cm. A B 3 extra parts = 21 cm 1 part = 7 cm
Ballpark Check & Math	B gets more than half of £96. £56 fits; $£40 + £56 = £96$.	Water is between 3 and 4 times the juice: between 54 and 72 litres. 63 fits.	The total is just over 4 times the 21 cm difference. $4 \times 21 = 84$; 91 is reasonable.
Final Answer	B receives £56.	63 litres of water.	91 cm altogether.

Ratio: check your method

Step-by-step practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method 12 parts; $\pounds 96 \div 12 = \pounds 8$ per part; $\pounds 8 \times 7 = \pounds 56$ for B.	Method 18 litres represent 2 parts. $18 \div 2 = 9$ per part; $9 \times 7 = 63$ litres of water.	Method The difference is $8 - 5 = 3$ parts. $21 \div 3 = 7$ cm per part; $7 \times 13 = 91$ cm.
If you got... $\pounds 40$: check whether you used $\pounds 8 \times 5$. That is A's share; the question asks for B.	If you got... 23 litres: check whether you used $18 + (7 - 2)$. Five extra parts do not mean five extra litres.	If you got... 273 cm: check whether you used 21×13 . The 21 cm represents 3 parts, not 1.
Check $\pounds 40 + \pounds 56 = \pounds 96$. B gets more than half of $\pounds 96$.	Check 18:63 simplifies to 2:7. Water is between 3 and 4 times the juice: 54 to 72 litres.	Check Lengths: 35 cm and 56 cm. Difference: 21 cm. Total: 91 cm.

Ratio: answers

Independent practice 1 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £10	1. 30 counters	1. £22
2. £36	2. 24 counters	2. £91
3. £24	3. 64 counters	3. £117
4. £30	4. 24 counters	4. £35
5. £18	5. 63 counters	5. £30
6. £49	6. 84 counters	6. £104

Check the amount requested: one share, the other share, or the total.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 1 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $1 + 3 = 4$ parts. $£40 \div 4 = £10$ per part. A gets $£10 \times 1 = £10$.	Method Q1: 10 red counters represent 1 part. Blue has 3 parts: $10 \times 3 = 30$ counters.	Method Q1: $5 - 2 = 3$ parts. $£33 \div 3 = £11$ per part. A gets $£11 \times 2 = £22$.
If you got... Q1 — £30: check whether you used $£10 \times 3$. That finds B's share, not A's.	If you got... Q1 — 12: check whether you used $10 + (3 - 1)$. Two extra parts mean 20 extra counters here.	If you got... Q1 — £66: check whether you used $£33 \times 2$. £33 represents 3 parts; divide by 3 first.
Check Q1: A £10 + B £30 = £40. The shares have ratio 1:3.	Check Q1: $10:30 = 1:3$. Blue has three times as many counters as red.	Check Q1: B gets £55; $£55 - £22 = £33$.

Ratio: answers

Independent practice 2 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £60	1. 24 counters	1. £65
2. £42	2. 56 counters	2. £135
3. £60	3. 110 counters	3. £55
4. £24	4. 64 counters	4. £36
5. £22	5. 77 counters	5. £168
6. £63	6. 108 counters	6. £80

Check the amount requested: one share, the other share, or the total.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 2 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $2 + 5 = 7$ parts. $£84 \div 7 = £12$ per part. B gets $£12 \times 5 = £60$.	Method Q1: 60 blue counters represent 5 parts. $60 \div 5 = 12$; red has $12 \times 2 = 24$.	Method Q1: $5 - 3 = 2$ parts. $£26 \div 2 = £13$ per part. B gets $£13 \times 5 = £65$.
If you got... Q1 — £24: check whether you used $£12 \times 2$. That is A's share; B has 5 parts.	If you got... Q1 — 57: check whether you used $60 - (5 - 2)$. Three fewer parts do not mean three fewer counters.	If you got... Q1 — £130: check whether you used $£26 \times 5$. The £26 difference represents 2 parts.
Check Q1: $£24 + £60 = £84$, with ratio 2:5.	Check Q1: $24:60 = 2:5$. Red has fewer counters than blue.	Check Q1: A gets £39. $£65 - £39 = £26$.

Ratio: answers

Independent practice 3 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £56	1. 98 counters	1. £165
2. £48	2. 80 counters	2. £51
3. £24	3. 132 counters	3. £117
4. £80	4. 70 counters	4. £143
5. £117	5. 52 counters	5. £126
6. £99	6. 154 counters	6. £132

Check the amount requested: one share, the other share, or the total.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 3 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $4 + 7 = 11$ parts. $£154 \div 11 = £14$ per part. A gets $£14 \times 4 = £56$.	Method Q1: 56 red counters represent 4 parts. $56 \div 4 = 14$; blue has $14 \times 7 = 98$.	Method Q1: $7 - 4 = 3$ parts. $£45 \div 3 = £15$ per part. Total: $£15 \times 11 = £165$.
If you got... Q1 — £98: check whether you used $£14 \times 7$. That finds B's share, not A's.	If you got... Q1 — 59: check whether you used $56 + (7 - 4)$. Three extra parts represent 42 counters here.	If you got... Q1 — £495: check whether you used $£45 \times 11$. £45 is the difference of 3 parts, not 1 part.
Check Q1: $£56 + £98 = £154$, with ratio 4:7.	Check Q1: $56:98 = 4:7$. Both amounts give 14 counters per part.	Check Q1: Shares £60 and £105; difference £45, total £165.

Ratio: answers

Independent practice 4 • Match the thread and question number.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
1. £112	1. 48 counters	1. £85
2. £108	2. 90 counters	2. £171
3. £126	3. 182 counters	3. £285
4. £84	4. 36 counters	4. £117
5. £75	5. 120 counters	5. £112
6. £117	6. 208 counters	6. £238

Check the amount requested: one share, the other share, or the total.

The next slide checks a method and a possible error in each thread.

Ratio: check your method

Independent practice 4 • Compare your working; correct the first wrong step.

Thread 1: Guided	Thread 2: Core	Thread 3: Depth
Method Q1: $3 + 7 = 10$ parts. $£160 \div 10 = £16$ per part. B gets $£16 \times 7 = £112$.	Method Q1: 112 blue counters represent 7 parts. $112 \div 7 = 16$; red has $16 \times 3 = 48$.	Method Q1: $7 - 5 = 2$ parts. $£34 \div 2 = £17$ per part. A gets $£17 \times 5 = £85$.
If you got... Q1 — £48: check whether you used $£16 \times 3$. That is A's share; the question asks for B.	If you got... Q1 — 108: check whether you used $112 - (7 - 3)$. Four fewer parts do not mean four fewer counters.	If you got... Q1 — £170: check whether you used $£34 \times 5$. £34 represents 2 parts; divide by 2 first.
Check Q1: $£48 + £112 = £160$, with ratio 3:7.	Check Q1: $48:112 = 3:7$. Red is less than half of blue.	Check Q1: B gets £119. $£119 - £85 = £34$.